

Quantum Tunneling-inspired Salp Swarm Algorithm (QT-SSA) for resilient coalition of point clouds

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Abstract - The Salp Swarm Algorithm (SSA) has evolved into a lightweight yet efficient metaphor influenced by salps' communal flocking behavior, delivering a viable approach for diverse optimistic problems. Nonetheless, SSA oftentimes exhibit minimal exploratory potential and endure a tendency to precocious convergence when employed to highly nonlinear and intricate search spaces such as Point Cloud Registration (PCR) alignment. A refined SSA variant, called quantum tunneling SSA (QT-SSA), is proposed as a way forward to these downsides. Initially, the baseline salp populace is applied with the quantum states, successively followed by the tunneling process to probabilistically prevail over the local bottlenecks and explore a broader search space. Moreover, the suggested QT-SSA is being assessed in aligning the actual-life 3D registration constraint involving point clouds that are quite stochastic and partly overlay. From the conducted assessments, it can be claimed that the proposed QT-SSA is a trustworthy and reliable optimization process that features noteworthy benefits over the typical method for addressing challenging PCR in the automated vision tasks.

Keyword - Salp Swarm Algorithm, Quantum Principles, Tunneling Mechanism, Point Cloud Registration, Transformation

I. INTRODUCTION

The optimization serves as a straightforward approach to identify the feasible solution in the sectors of science and engineering to handle the optimistic queries [1]. It implies the refinement of the potential solution to optimize the intent function within an identified constraints environment [2]. Classical optimization algorithms (OA), including the gradient-based technique, typically experience challenges in large-scale, uncertain search domains and tend to get up in confined minima [3]. As a potentiality to these shortcomings, nature-inspired metaheuristic OA have evolved into a resilient and efficacious approach in the quest for global optima [4]. The SSA has been exceptionally potent in resolving the challenging aspect related to optimization [5]. The SSA is deployed on the prominent-admirer modalities, imitating the swarming routine of salps in the waterways [6]. Regardless of its upsides, the standard SSA's productivity in complex dimensions and multimodal

tasks is impeded by its convergent setback and local confinements [7-8]. To address the confines of SSA, this study presents a quantum mechanics-inspired SSA named QT-SSA. The QT states that the particles can penetrate the energy hurdles that would be impenetrable in classical physics through the quantum phenomena [9-10]. The suggested QT-SSA features stability in exploration and exploitation, by facilitating the particles to get away from the confinements of the probabilistic impediment penetration. This quantum-enabled framework drives up the search variance and streamlines the convergent process towards the optimum, ensuring the suggested QT-SSA, an intriguing approach for real-life scenarios. In machine vision, automation and 3D restoration, PCR is a vital procedure wherein distinct 3D dynamic point sets necessitate to be precisely oriented to yield a unified and coherent visualization of an object or scene [11]. Classical registration approaches, including the Iterative Closest Point (ICP) and Coherent Point Drift (CPD) are susceptible to substantial processing overhead, vulnerable to initial constraints and confined to minima convergence [12]. By adequately optimizing the transformation parameters, the metaphor-based swarm-inspired approaches have been explored to boost the registration process [13]. The QT mechanism leverages the standard SSA by preventing sub-optimal convergence and local minima sloth [14]. Employing the suggested QT-SSA for PCR suggests an innovative optimization-driven framework that strikes the exploration-exploitation harmony in the parameter search domain, assuring trustworthy and realistic point cloud alignment [15].

The subsequent sections of the study are dispersed as follows: Section 2 presents an in-depth review of standard SSA and PCR available in literature. The thorough explanation of the standard SSA, quantum mechanics and tunneling process is outlined in Section 3. In Section 4, the suggested QT-SSA approach is delivered. Section 5 covers the optimality of the proposed QT-SSA in the readily apparent of registration process. The final assessment,

alongside the future prospects of the work, is reported in Section 6.

II. RELATED WORK

This segment discusses the pros and cons related to the standard SSA, quantum mechanics, tunnelling and its application in PCR available in the literature. Mahapatra et al. introduced a hybrid Quantized Orthogonal Experimentation SSA (QOA-SSA) to advance the population diversity and global convergence. The authors had assessed their QOA-SSA model against the IEEE-CEC-14 benchmark suite. The authors had also contrasted their QOA-SSA model with present-day methods, including WOA, SCA, MFO, FFO and CSA. The non-parametric analyses are also adopted to reliably verify the QOA-SSA [16]. Mahmoudi et al. pitched a Quantum-inspired PSO clustering (QIPSOC) for lowering the consumption of energy in substantial IoT networks. The authors had developed MPMC for deploying the constraints for IoT networks such as multi-hop routing, flexible base station location and cluster energy limits [17]. Naik et al. devised a QIPSO for workflow planning (QIPSO-WSA) to take on the concerns of cost-effective task scheduling in modern computing scenarios. The authors' QIPSO-WSA model is a well-balanced variant of standard PSO involving the quantum-driven strategy, multi-objective optimization and encoding-decoding framework [18]. Paul et al. introduced a QPSO paradigm for multiple-purpose handling of energy in a grid-linked mini-grid with the twin objectives of limiting expenses and mitigating pollutants. The authors resolved

the energy handling concern by the inclusion of clean energy sources, namely solar and wind [19]. Wang et al. addressed the shortcomings in resource description by refining the assessment of Young's modulus and Poisson's ration utilizing the improved QPSO (IQPSO). The author had adopted a few methods to IQPSO for its optimality, such as chaotic mapping for baseline population diversity, Levy's method to elude the stagnation from local minima and Gaussian mutation approach for dense accuracy [20]. Xu et al. developed QPSO strengthened with chaotic encoding schemes for Flexible Job-Shop Planning (FJSP) [21]. Yan et al. tackled a major challenge in the marine wind turbine servicing by employing an advanced Levy flight QPSO (ELQPSO) to boost the instant stress field inspection in offshore wind turbines [22]. Feng et al. developed the PCR-GW method, inspired by the Grey Wolf Optimizer, to address the drawbacks related to the reduced registration accuracy and extensive processing time. Their hybrid methodology comprises global optimization and data extraction [23]. Zhan et al. developed an innovation method referred to as Entropy-based PSO (EPSO) for robust PCR tasks. The authors had employed a k-d tree to develop a structure among the points to resolve the drawbacks of noise and inferior registration accuracy, simplifying the noise by employing a mean filtering approach [25]. Improving on the prior work, Zhan et al. created a sophisticated feature-driven PCR referred to as Normal Vector (NV) and PSO. Rather than focusing solely on the entire PCR, the authors had focused on precise PCR matching [24]. The in-depth review concerning the standard SSA, quantum mechanics, tunnelling and its application in PCR is outlined in Table 1.

TABLE I. LITERATURE CONCERNING THE STANDARD SSA, QUANTUM MECHANICS, TUNNELLING AND ITS APPLICATION IN PCR

Author	Methodology	Dataset/ Work	Year
Mahapatra et al. [16]	QOX-SSA	Feature selection and RBFNN training	2025
Mahmoudi et al. [17]	QIPSOC	To lower the energy consumption Substantial IoT network simulations	2025
Naik et al. [18]	QIPSO-WSA	Cost-effective task scheduling	2025
Paul et al. [19]	QPSO	For limiting expenses and mitigating pollutants in micro grid energy paradigm	2025
Wang et al. [20]	IQPSO	Circle chaotic mapping, Gaussian mutation and Levy flights	2025
Xu et al. [21]	QPSO	Chaotic Mapping and FJSP	2025
Yan et al. [22]	ELQPSO	Marine wind turbine servicing	2025
Feng et al. [23]	PCR-GW	Stanford 3D dataset -bunny, dragon, armadillo	2020
Zhan et al. [24]	Feature-based PCR utilizing PSO and NV	Stanford 3D dataset and custom synthetic	2020
Zhan et al. [25]	EPSO-PCR	Stanford 3D dataset- cow, man bunny	2018

III. BASIC PRELIMINARIES

This segment elaborates about the OA castoff and the mathematical operators employed namely quantum and tunneling.

A. SSA

The standard SSA was developed by Mirjalili et al. in 2017, driven by the remarkable and unique instincts of salps, a

species of marine creatures that mimic jellyfish and are commonly found in the oceans [6]. The salps normally cluster into a long chain-like structure when travelling and foraging, which permits them to navigate swiftly in pursuit of foodstuff. This chain development resembles a prominent-admirer salp dynamic, where the prominent salp lays the path towards optimality and admirer salps trailing suit [26-28]. By replicating the communal and adaptive

behavior of the salp in the ocean, the classical SSA intends to mimic the prominent-admirer salp dynamic, which may resolve the complicated practical-world tasks.

$$S_m = \begin{cases} T_m + V_1((C_m - F_m)V_2 + T_m)V_3 \geq 0 \\ T_m - V_1((C_m - F_m)V_2 + T_m)V_3 < 0 \end{cases} \quad (1)$$

The location of the prominent salp is signified as S_m . The location of the foodstuff source is denoted as T_m . The ceil and floor bounds are indicated as C_m and F_m . The random numbers that regulate the standard SSA are V_1 , V_2 and V_3 . The V_1 controls the balance between directed exploration and exploitation. Meanwhile, the V_2 and V_3 are the uniform random numbers between 0 and 1. b and B represent the present repetition and the total number of reiterations. The location of the m_{th} admirer salp at n_{th} dimension is signified as L_m^n .

$$V_1 = 2e^{-\left(\frac{b}{B}\right)^2} \quad (2)$$

$$L_m^n = \frac{(L_m^n + L_m^{n-1})}{2} \quad (3)$$

B. Quantum

In the quantum framework, the present state of the particle or a system is expressed as a quantum state vector, generally stated as a ket and defined using Dirac notation as:

$$|\sigma\rangle = \alpha|0\rangle + \beta|1\rangle \quad (4)$$

Where $|\sigma\rangle$ is the superposition state of the system, $|0\rangle$ and $|1\rangle$ are the reference or basis states. The transition amplitudes to occur in states of $|0\rangle$ and $|1\rangle$ are outlined by the complex numbers α and β . $|\alpha|^2$ and $|\beta|^2$ ensure the likelihood of being in $|0\rangle$ and $|1\rangle$. The overall likelihood is kept intact and presented in Eq. 5. The quantum notation enables the particles to explore broadly in the search space by evading the confined minima and develop better exploration [29].

$$|\alpha|^2 + |\beta|^2 = 1 \quad (5)$$

C. Tunneling

The search agents subsequently proceed to the tunneling procedure, during which the agents are tunneled into the fresh sectors of the search domain regulated by a tunneling likelihood parameter γ on the quantum state $|\sigma\rangle$ [30]. The value of parameter γ declines over the reiterations. The fresh computed location of m_{th} salp is represented as NL_m , and the tunneling perturbation of m_{th} salp is denoted as ΔL_m . The *rand* signifies a random integer in the assortment of 0 and 1. The initial likelihood is denoted as γ_0 .

$$NL_m = L_m + \Delta L_m \quad (6)$$

$$\Delta L_m = (\alpha - \beta) \cdot \gamma \cdot (rand - 0.5)(C_m - F_m) \quad (7)$$

$$\gamma = \gamma_0 \cdot \left(1 - \frac{b}{B}\right) \quad (8)$$

IV. PROPOSED MODEL

The subsequent segment delivers an in-depth account of the framework and the conceptual underpinnings of the suggested QT-SSA approach. The approach builds upon the latent of standard SSA and employs the quantum tunneling concept to fortify the exploratory potential. The underlying intent of the transformation is to enhance the metaphor OA's trait to traverse intricate search areas reliably and to avert the untimely convergence to less-than-ideal solutions. In metaphor optimization, the attainment of an approach significantly hinges on its power to strike a balanced exploration-exploitation.

Quantum mechanics is applied to the process of initialization. The initial salps are signified employing the quantum-driven states, in which the location of each salp is defined by a likelihood distribution rather than an explicit definition. This likelihood representation assists a broader baseline coverage of the search space. Furthermore, tunneling processes are incorporated into the quantum SSA to maintain its consistency. This technique empowers some contender salps to go before high-energy barriers to conceivably better regions in the search domain by probabilistically performing a tunneling jump instead of following traditional movement patterns. This mechanism is carried out by randomly perturbing the solution distant from the local neighborhood about a tunneling likelihood factor, which facilitates the salp to escape confined minima and to discover uncharted territory. The combined working of quantum principles and the tunneling mechanism enriches the diversity of the initial population and streamlines the convergence by driving intense early-stage exploration. The salps adjust their locations throughout the process in each reiteration by following the frontrunner salp or by enduring the tunneling process. The detailed table on the parameter setting is defined in Table 3.

In the context of PCR, QT-SSA is employed to optimize the rigid transformation parameters, namely rotation and translation, that optimally align the source and target point clouds. The transformation parameters are signified as a set of salps in the search space. The fitness of each contender salp is assessed based on the utility function, which is the RMSE between the transformed source points and the target points. The inclusion of the suggested QT-SSA approach to the registration process enables it to attain better accuracy, higher resilience against initial misalignments and quicker convergence towards optimality. The algorithmic representation and detailed workflow of QT-SSA for PCR is presented in Table 2 and Fig. 1 respectively.

TABLE II. PSEUDOCODE REPRESENTATION OF QT-SSA FOR PCR

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Algorithm: QT-SSA
Initialize the baseline population  $S_m$  for each salp  $m$  based on  $C$  and  $F$ 
Initialize the quantum state  $q$  for each salp  $m$ 
for each iteration  $q = 1$  to  $Q$  do
    Compute the fitness of each salp utilizing the utility function
    Allocate the optimum salp to  $T$ 
    Alter the  $V_1$  value by Eq. 2
    for every salp  $S_m$  search agent do
        if  $S_m$  is, prominent salp, then
            Adjust prominent salp location by employing Eq. 1
        otherwise
            Adjust the admirer salp location by employing Eq. 3
        end if
        Update the location  $NL_m$  of each salp  $m$  by tunnelling operation employing Eq. 7
        if  $NL_m$  is better than  $S_m$ 
            Alter the prominent salp  $S_m$ 
        Adjust the location of contender salps employing  $C$  and  $F$ 
    end for
end for
return  $T$ 
end QT-SSA

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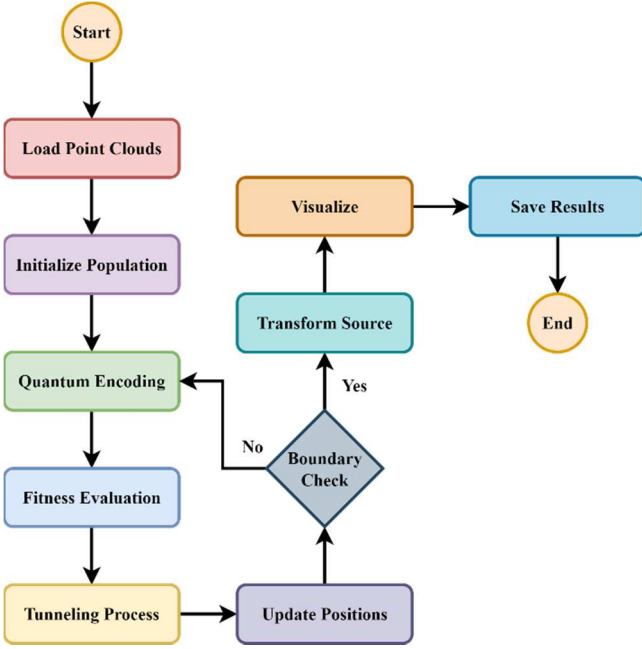


Fig. 1. Flow graph representation of the QT-SSA for PCR

TABLE III. PARAMETER SETTINGS

Parameter	Symbol	Value
Baseline population size (Number of salps)	N	30
Repetition count	I	100
Problem dimension	D	6
		three rotational (R_x, R_y, R_z) and three translational (T_x, T_y, T_z)
Tunneling likelihood	γ	0.6 decays linearly
Confined bounds	$[F, C]$	Rotational: $[-\pi, \pi]$ and Translational: $[-1, 1]$

V. RESULTS AND DISCUSSION

In the present study, the suggested QT-SSA approach has been utilized to assess the registration process utilizing the Bildstein and Castleblatten Station1 dataset [31]. The dataset comprises the dense 3D laser-scanned points that duplicate actual-life settings. The intent objective of this study is to reallocate the source point cloud onto the target point cloud to pinpoint the optimum rigid transformation. The transformation is governed by six crucial parameters: three rotation parameters (R_x, R_y, R_z) and three translation parameters (T_x, T_y, T_z) . The rotation parameters facilitate rectifying the source point clouds' location by specifying the angular orientations concerning the x, y , and z axes. The translation parameters simplify the spatial repositioning of the source point clouds about the shift along the x, y , and z axes. The baseline 3D spatial points are referred to as the source point clouds that need to be registered. The optimum rotation and translation parameters that are applied to the source result in the transformed target points. The search bounds for rotation and translations are defined in $[-\pi, \pi]$ and $[-1, 1]$. The utility function is to minimize the mean distance between the source and target points. The optimal computed parameters that are acquired throughout the registration process are outlined in Table 4. The transformation of the source point over the reiterations is represented in Fig. 2. The runtime complexity of the suggested QT-SSA is computed to be $O(N \cdot I \cdot D)$, where N is the count of salps particle considered, I is the repetition count and the D is the problem dimension. The outcomes unequivocally highlight the extent to which the suggested QT-SSA approach performs in minimizing the registration errors in contrast with the classical registration method ICP and CPD along with standard SSA.

TABLE IV. COMPUTED OPTIMAL TRANSFORMATION PARAMETERS

Dataset	Bildstein Station1				Castleblatten Station1			
Parameter	ICP	CPD	SSA	QT-SSA	ICP	CPD	SSA	QT-SSA
R_x	3.6500	4.5216	3.4137	2.9481	4.5411	5.5331	4.4646	5.7352
R_y	-0.4702	-0.4720	-0.4230	-0.2775	-0.4816	-0.4234	-0.4341	-0.4562
R_z	-0.7723	-0.6604	-0.9141	-0.5047	-0.5558	-0.6631	-0.6449	-0.5951
T_x	1.3115	1.3096	1.2912	0.8345	1.4891	1.5243	1.1497	1.6254
T_y	0.8091	1.1372	0.9649	0.7752	1.2703	0.8919	1.1957	1.2988
T_z	-1.1871	-1.0150	-1.1338	-0.6996	-0.8349	-1.1366	-1.0677	-1.3419
Best Fitness Value	0.4614	0.4556	0.4248	0.2676	0.4498	0.5252	0.4503	0.2656
Accuracy	82.708	85.075	94.807	97.850	85.735	88.342	94.673	98.705
Precision	81.256	84.427	94.455	96.580	84.709	87.440	93.429	98.198
Sensitivity	80.074	84.877	93.602	96.870	84.492	86.980	93.052	97.423
Specificity	80.437	84.786	93.848	96.920	84.150	86.721	93.118	97.478
Min RMSE	7.8820	6.4259	7.7681	4.4910	8.3635	8.4195	4.4489	3.8304
Standard Deviation	6.4153	6.8380	5.7211	3.4788	4.2897	6.0738	6.4474	2.0439
Average Error	5.1995	4.4601	4.8536	2.8403	4.9770	4.0048	5.4307	2.6998

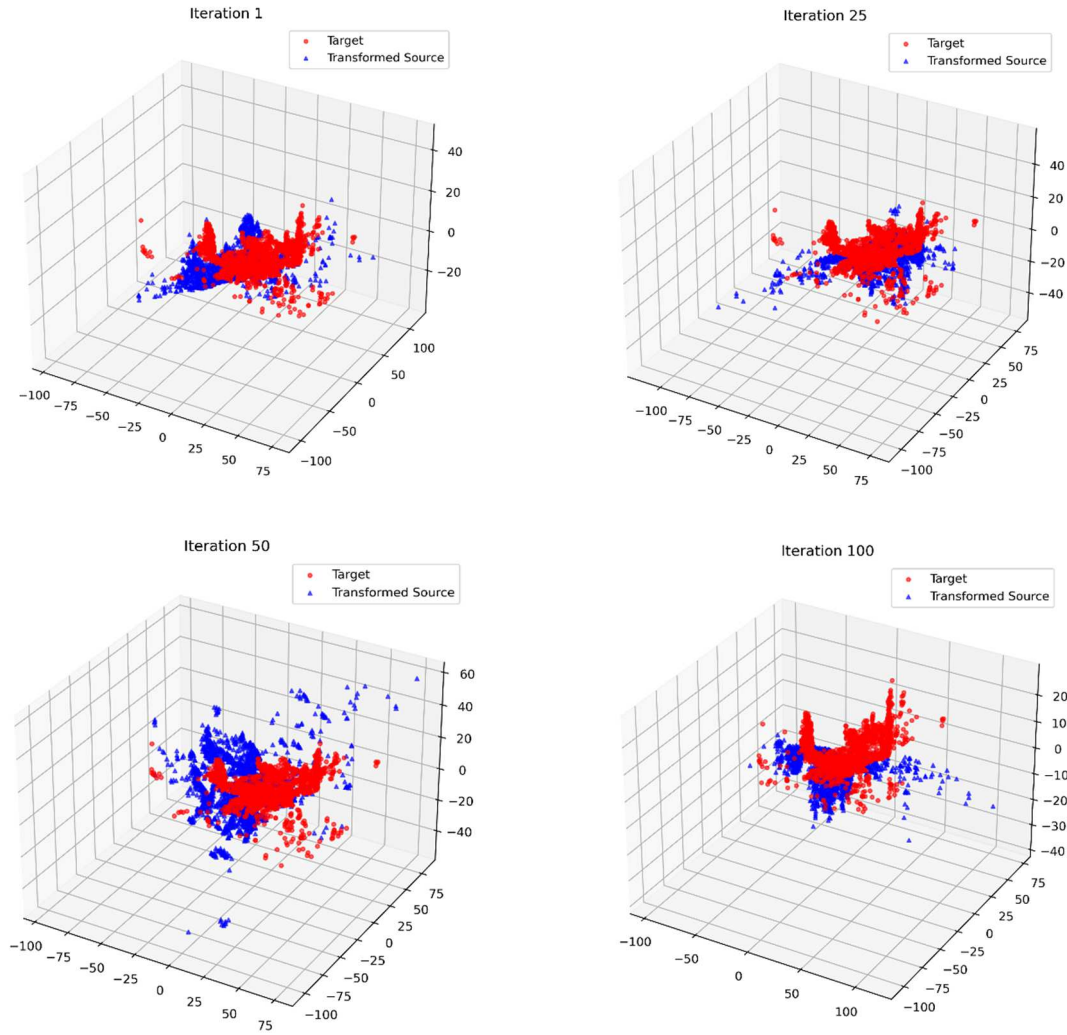


Fig. 2 Transformation of source points over the reiterations

VI. CONCLUSION AND FUTURE SCOPE

In this study, we have developed a fresh technique termed QT-SSA for the coalition of the point clouds. Initially, the standard SSA has been considered due to the simpleness,

ease of implementation and intensive exploration trait. However, the standard SSA may end up in the confined minima, particularly in intricate, multifaceted and registration processes. To overcome the shortcomings of SSA, we unified quantum mechanics with the tunneling

process to probabilistically escape the confined stagnation and to enable fresh regions of exploration with a broader range of solution suites. Henceforth, the prowess of the QT-SSA has been greatly enhanced with a well-stabilized exploration-exploitation. Subsequently, the proposed QT-SSA approach is deployed to align the point clouds in the registration process. The baseline points are taken from the Bildstein and Castleblatten Station1 dataset, and an attempt has been made to align them with the target points with the optimized set of transformation parameters. The findings from the study showcase that the suggested QT-SSA is a reliable and trustworthy method for the coalition of the point clouds. Although the proposed QT-SSA performed with remarkable proficiency, further studies and developments are still being made. Firstly, the suggested QT-SSA approach exhibits a slightly higher computing overhead when benchmarked with lightweight registration algorithms such as ICP and CPD. Secondly, the proposed QT-SSA may require model adaptation when applied to the task of non-rigid or deformable registration. Addressing these aspects is a probable pathway for subsequent research.

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